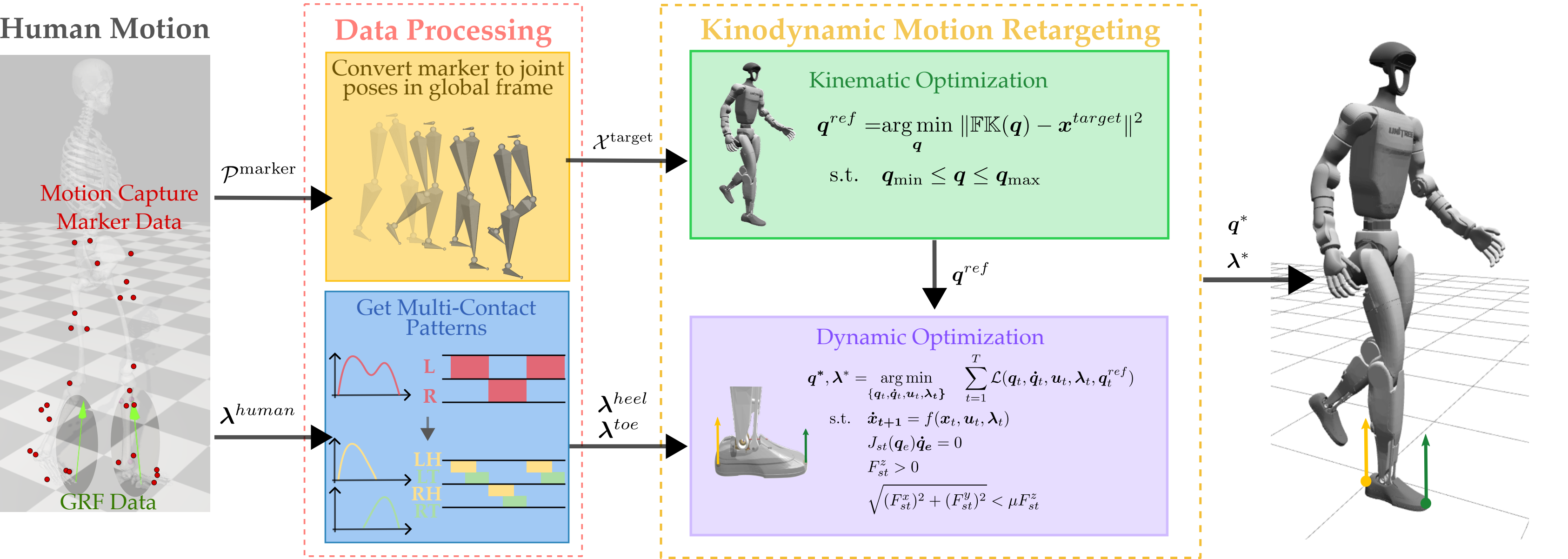

Human Motion
Motion Capture Marker Data
GRF Data
$\mathcal{P}^{\text{marker}}$
$\boldsymbol{\lambda}^{human}$
Data Processing
Convert marker to joint poses in global frame
Get Multi-Contact Patterns
L
R
LH
LT
RH
RT
$\mathcal{X}^{\text{target}}$
$\boldsymbol{\lambda}^{heel}$
$\boldsymbol{\lambda}^{toe}$
Kinodynamic Motion Retargeting
Kinematic Optimization
$\boldsymbol{q}^{ref} = \arg\min_{\boldsymbol{q}} \|\mathbb{FK}(\boldsymbol{q}) - \boldsymbol{x}^{target}\|^2$
s.t. $\boldsymbol{q}_{\min} \leq \boldsymbol{q} \leq \boldsymbol{q}_{\max}$
$\boldsymbol{q}^{ref}$
Dynamic Optimization
$\boldsymbol{q}^*, \boldsymbol{\lambda}^* = \underset{\{\boldsymbol{q}_t, \dot{\boldsymbol{q}}_t, \boldsymbol{u}_t, \boldsymbol{\lambda}_t\}}{\arg\min} \sum_{t=1}^{T} \mathcal{L}(\boldsymbol{q}_t, \dot{\boldsymbol{q}}_t, \boldsymbol{u}_t, \boldsymbol{\lambda}_t, \boldsymbol{q}_t^{ref})$
s.t. $\dot{\boldsymbol{x}}_{t+1} = f(\boldsymbol{x}_t, \boldsymbol{u}_t, \boldsymbol{\lambda}_t)$
$J_{st}(\boldsymbol{q}_e)\dot{\boldsymbol{q}}_e = 0$
$F_{st}^z > 0$
$\sqrt{(F_{st}^x)^2 + (F_{st}^y)^2} < \mu F_{st}^z$
$\boldsymbol{q}^*$
$\boldsymbol{\lambda}^*$
UNITREE